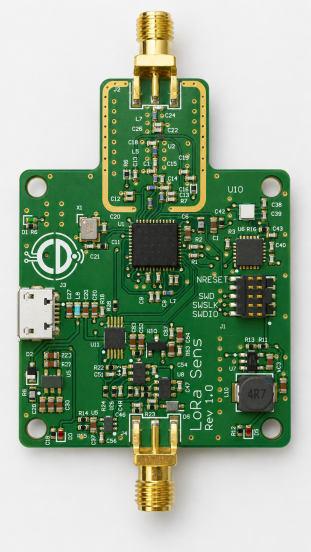
PCB, IoT Solutions & Embedded Engineering Portfolio

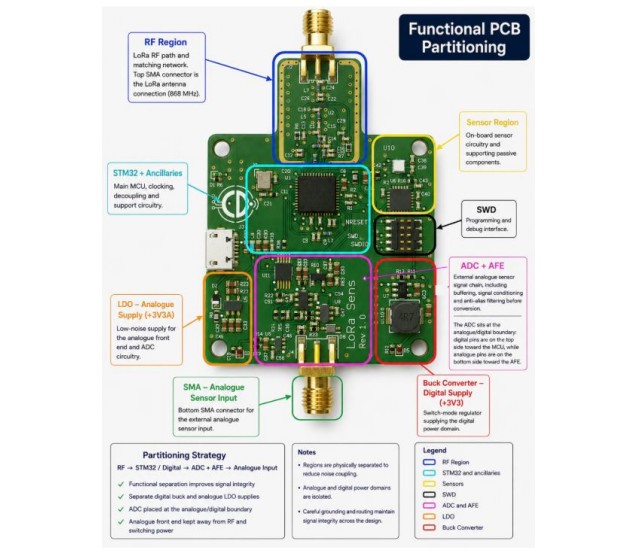
Featured project: STM32WLE5 LoRa Sensor Node | mixed-signal IoT | sub-GHz RF | 4-layer PCB

Embedded hardware design and review from schematic through PCB layout, verification planning and manufacturing outputs - practical electronics aimed at manufacturable, supportable products.

STM32WL PCB / assembly view



Functional PCB partitioning

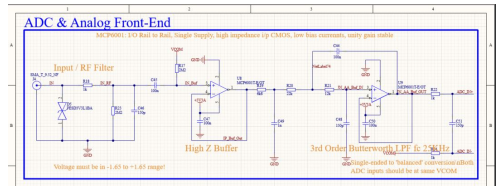


Design intent

STM32 + LoRa / RF

Analog + digital + sensors

The board deliberately separates RF, MCU/digital, ADC/AFE and power functions so the physical layout reflects the system architecture.



FEATURED PROJECT - DESIGN HIGHLIGHTS

- STM32WL embedded + LoRa / sub-GHz RF architecture
- RF matching / filtering and antenna interface
- Analog front end, SAR ADC and sensor integration
- Separate analog / digital power approaches

WHAT I DELIVER

- **Schematic & PCB design / review**
- **RF, mixed-signal & embedded hardware**
- **DFM / DFT and release documentation**
- **Bring-up, debug & verification support**

TOOLS / WORKFLOW

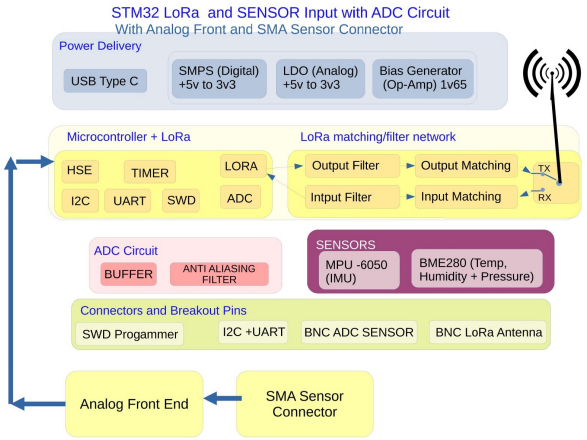
- **Altium Designer**
- **LTspice + Python/Jupyter**
- **DRC + fabrication outputs**
- **EVT / DVT / PVT planning**

SECTORS Industrial | Medical Devices | Environmental Monitoring | Energy | Consumer / Connected Products

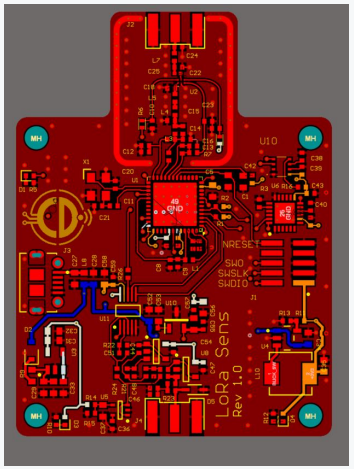
STM32WL Architecture, PCB & Review Evidence

System-level design evidence rather than a theory-heavy report: architecture, layout, signal chain and physical partitioning.

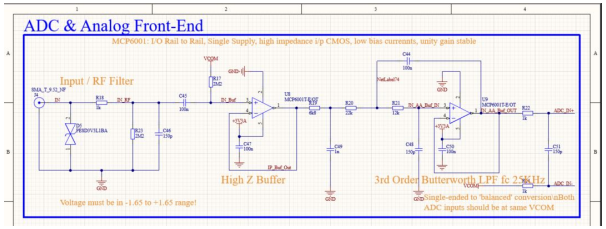
System architecture



PCB layout

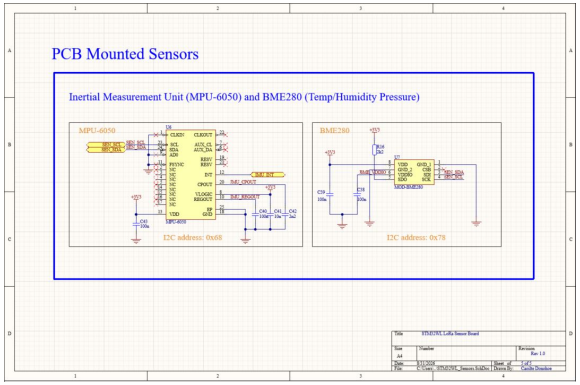


Analog front end + ADC



- Input protection / filtering
- High-Z buffering
- Anti-alias filtering
- SAR ADC interface

Sensors + embedded interfaces



- I2C sensors
- SWD / debug access
- MCU peripheral interfaces
- Board-level integration

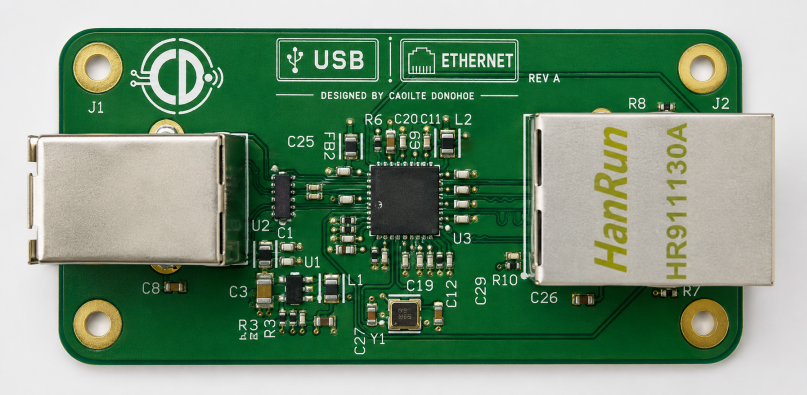
Design review lens

- Grounding and return paths
- Filtering / decoupling / protection
- RF placement and matching network
- Analog / digital power separation
- ADC boundary placement
- Test access and debug interfaces
- Manufacturing / certification-risk review

High-Speed PCB Design to Manufacturing Release

USB 5 Gbit/s to Gigabit Ethernet Adapter - a second project showing controlled routing, impedance planning and release documentation.

LAN7800 USB-to-Ethernet PCB



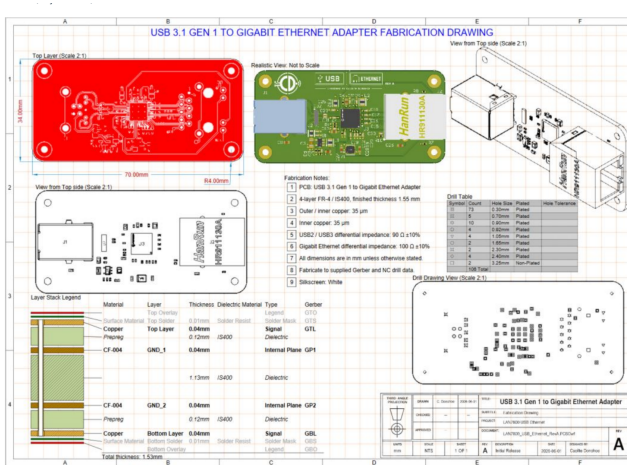
Project summary

USB 5 Gbit/s -> Gigabit Ethernet

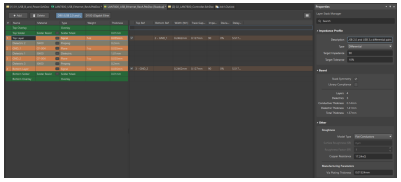
4-layer LAN7800 adapter designed from schematic capture through PCB layout and fabrication outputs.

- Controlled USB and Ethernet differential routing
- Layer-stack and impedance-profile planning
- Component integration and rule-driven DRC
- Gerber X2, NC Drill and editable source

Fabrication drawing / release evidence



Layer stack & impedance profile



Release package

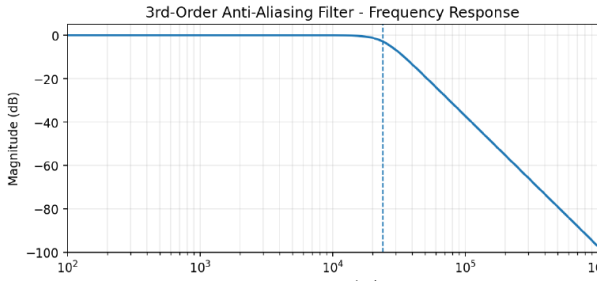
Schematic	editable design + PDF
PCB	routed 4-layer layout
BOM	manufacturer + MPN
Fabrication	Gerber X2 + NC Drill
Verification	final DRC

Verification, Simulation & Supporting Engineering

Simulation supports design decisions; the portfolio remains focused on real PCB, embedded, RF and production-readiness work.

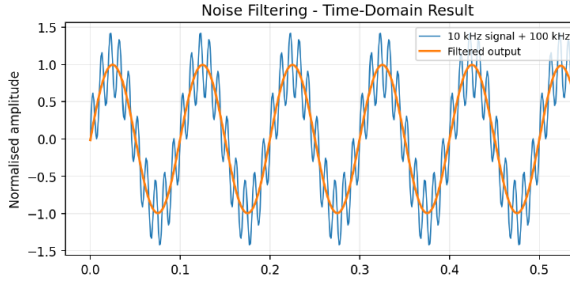
Analog filter response

Analog filter response



Noise filtering - time domain

Noise filtering - time domain



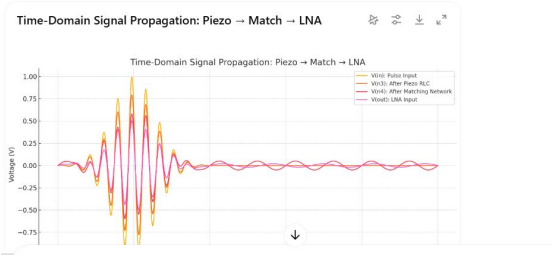
STM32WL verification

- Sallen-Key anti-alias filter analysis
- Bias / VCOM and ADC signal-path checks
- Transient and pulse-response analysis
- 868 MHz RF matching / filter trade-offs
- Python/Jupyter frequency and time-domain checks

Additional support

Board bring-up | firmware integration | hardware debug | DFM / DFT | EMC / pre-compliance risk review

Ultrasonic AFE - small supporting study



5.4 MHz piezo -> matching network -> LNA analysis looking at resonance, loading and time-domain transfer. Included as a small flavour of related analogue-front-end work.

Production-readiness review

- Schematic and multilayer PCB review
- Component selection / compatibility
- Power, battery and low-power design
- RF / antenna / EMC risk review
- Grounding, filtering and protection
- DFM / DFT and manufacturing outputs
- Bring-up and validation planning